DATA SCIENCE & ANALYTICS · BUSINESS INTELLIGENCE REPORT

Project FORESIGHT

Demand & Inventory Intelligence for NorthBay Living
A Detailed Business Intelligence & Analytics Report

617M	2M	304.93	200
Total Revenue (INR)	Units Sold	Avg. Unit Price (INR)	Active SKUs

PREPARED BY

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Client: **NorthBay Living** | Program: Zidio Development — Data Science & Analytics Internship
Source: Power BI Desktop dashboard export · Reporting window: FY2021–FY2026 (FY2026 partial)
Document v1.0 · Confidential — prepared for internal client review

CONTENTS

Table of Contents

This report expands the Project FORESIGHT Power BI dashboard into a detailed, narrative business-intelligence document with commentary, tables and analyst notes for every dashboard section.

FRONT MATTER

Document Control & About This Report

Field	Detail
Document title	Project FORESIGHT — Detailed BI & Analytics Report
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How to Read This Report

This document expands the twelve-section Power BI dashboard produced for the FORESIGHT engagement into a detailed narrative report. Each dashboard page is treated here as its own chapter: the underlying figures are reproduced in table form, accompanied by written commentary that explains what the numbers show, why it matters to NorthBay Living's stocking and merchandising decisions, and where the data itself deserves a second look before being used to drive strategy.

Chapters 1–12 mirror the dashboard structure exactly, in the same order the Head of Operations would click through it. Chapter 13 consolidates every data-quality caveat raised along the way into a single reference list. Chapters 14 and 15 step back from the individual charts to synthesize the findings into a small number of insights and a concrete action plan. Chapter 16 and the appendices document how the report was assembled, the definitions behind each metric, and the underlying data schema, so that the analysis can be audited, refreshed, or handed to another analyst without re-deriving it from scratch.

METHOD NOTE — How figures are sourced

All figures in Chapters 1–12 are taken directly from the Power BI export unless otherwise flagged. Where the source dashboard did not publish an exact value (for example, individual bar values on the Top/Bottom SKU charts), this report says so explicitly rather than presenting an invented number as fact.

EXECUTIVE SUMMARY

What This Report Found

A condensed view of the analysis for readers who need the headline story before the detail.

NorthBay Living generated **INR 617M** in revenue across **2M units sold** and **200 active SKUs** over the FY2021–FY2026 reporting window, at an average realized unit price of **INR 304.93**. The business is heavily concentrated: Furniture alone accounts for 78% of revenue despite representing 43% of the SKU catalog, and a small number of legacy SKUs launched in 2019 and 2021 still generate roughly 70% of total revenue.

The analysis surfaces five findings with direct operational relevance. First, Furniture is simultaneously the highest-revenue, highest-price and highest-stockout-risk category (43% of item-days out of stock), making it the clear priority for any reorder-planning system. Second, promotions are not a marginal tactic at NorthBay — 62–65% of every category's revenue is earned on promoted SKU-days, meaning promotional activity should be modelled as a baseline demand driver rather than a special event. Third, the holiday flag in the current dataset shows no measurable effect on daily revenue (a 0.06% gap between holiday and non-holiday days), so it should not be relied upon as a demand signal without further investigation. Fourth, on-hand inventory roughly doubled between 2021 and 2023 without a corresponding fall in stockout rates, pointing to a stocking-mix problem rather than a stocking-volume problem. Fifth, and most importantly for how this data should be used going forward, revenue, unit sales and stockout counts are distributed almost perfectly evenly across all four sales regions — a pattern real retail data essentially never produces, and one that should be validated at the source before any region-specific strategy is built on top of it.

Summary Scorecard

617M	2M	304.93	200
Total Revenue (INR)	Units Sold	Avg. Unit Price (INR)	Active SKUs

Top-Line Findings at a Glance

- **Furniture dominates the business:** 78% of revenue, the highest unit price (INR 544), and the highest stockout rate (43%) of the three categories.
- **Promotions are the revenue engine, not a lever:** 62–65% of category revenue is earned on promoted SKU-days across Furniture, Decor and Small Appliances alike.
- **The holiday flag carries no signal:** average daily revenue is effectively identical on holiday (INR 1,688.43) and non-holiday (INR 1,689.51) days.
- **Inventory grew faster than stockouts fell:** on-hand units nearly doubled from 2021 to 2023, yet Furniture and Decor still stock out on 39–43% of item-days.
- **Regional and stockout splits are suspiciously even:** East, North, South and West each sit within roughly a quarter-point of 25% of revenue, units and stockout counts — a data-quality flag, not a business insight, until validated.
- **Revenue is legacy-driven:** SKUs launched in 2019 and 2021 alone contribute close to 70% of total revenue.

Reading Guide by Stakeholder

Different readers will want different parts of this report. The table below is a quick routing guide for the Head of Operations, the merchandising team and the finance lead, matching NorthBay's own stated stakeholder list from the engagement brief.

Reader	Recommended chapters
Head of Operations	Chapters 1, 10, 13, 14, 15 — stockout risk, inventory position, and the recommended action plan.
Merchandiser	Chapters 3, 4, 8, 11, 12 — category and subcategory performance, promotion effectiveness, and SKU-level detail.
Finance lead	Chapters 1, 9, 14 — headline revenue, pricing and discount behaviour, and the rupee-relevant findings in the synthesis chapter.
Data / analytics team	Chapters 13 and 16, plus Appendices A–B — data-quality flags, methodology, and the metric and schema reference.

The remainder of this report proceeds section by section through the dashboard, in the same order it is presented in Power BI, before consolidating everything into the synthesis and recommendations chapters near the end.

CHAPTER 01

Company-Wide Performance Overview

Revenue, units and SKU footprint across the full analysis window — the dashboard's landing page.

617M Total Revenue (INR)	**2M** Units Sold	**304.93** Avg. Unit Price (INR)	**200** Active SKUs

The Executive Overview page is the first thing any stakeholder sees, and it is designed to answer a single question in under five seconds: is the business healthy? At a headline level the answer is qualified yes — INR 617M in revenue across 200 SKUs at a stable average price near INR 305 — but every KPI on this page has a more interesting story underneath it once the category and regional splits are examined in later chapters.

Yearly Revenue Trend

Revenue grew steadily from an estimated INR 38M in 2021 to a peak of roughly INR 148M in 2025, before falling to approximately INR 79M in 2026. This drop is not a business decline — 2026 is a partial year in the source data (the export was taken mid-year) — but it will visually read as a collapse to anyone glancing at the chart without that context. Any client-facing version of this chart should carry an explicit "2026 year-to-date" annotation, or exclude the incomplete year from the trend line entirely, to avoid a false impression of decline.

Year	Revenue (INR M, approx.)
2021	38
2022	112
2023	118
2024	122
2025	148
2026*	79

**2026 figures reflect a partial year and are not directly comparable to full-year totals.*

Revenue and Units by Category

Furniture contributes INR 0.48bn of the INR 0.617bn total (78%), with Small Appliances and Decor essentially tied at INR 0.07bn each (11% and 11%). The pattern repeats in units sold: Furniture leads with 0.88M units, Decor follows at 0.73M, and Small Appliances trails at 0.40M. Because Furniture also carries the highest average unit price (detailed in Chapter 3), its revenue share is disproportionately larger than its unit-volume share — a hallmark of a premium, considered-purchase category anchoring the business.

Category	Revenue	Units Sold
Furniture	0.48bn (78.1%)	0.88M (43.6%)
Small Appliances	0.07bn (11.3%)	0.40M (19.8%)
Decor	0.07bn (10.6%)	0.73M (36.6%)

Revenue Share by Region

Revenue splits almost exactly evenly across the four sales regions: East (INR 155M, 25.07%), North (INR 154M, 25.05%), West (INR 154M, 24.96%) and South (INR 154M, 24.93%). Each region sits within a 0.14-percentage-point band of a perfect quarter share.

Region	Revenue	Share of Total
East	INR 155M	25.07%
North	INR 154M	25.05%
West	INR 154M	24.96%
South	INR 154M	24.93%

DATA QUALITY FLAG — An unusually even regional split

Real-world regional demand is shaped by population density, store or delivery coverage, local competition and seasonality by climate — it essentially never lands within a quarter of a point of an equal four-way split. This pattern recurs on the Regional Performance page (Chapter 5) and the Inventory & Stockout Analysis page (Chapter 10). We recommend the analytics team confirm the region field's join logic in the underlying sales_daily / calendar tables before any region-specific stocking or marketing decision is made on this basis.

Taken together, the Executive Overview establishes three themes that recur throughout this report: Furniture's outsized importance to the business, the partial-year distortion in the 2026 figures, and a regional data pattern that warrants validation. Each is revisited with more depth in the chapters that follow.

CHAPTER 02

Sales Trend & Time Intelligence

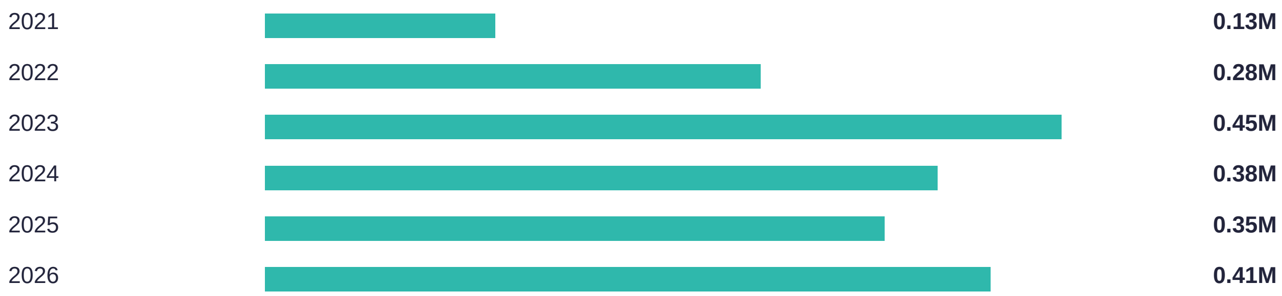
Revenue and unit-sales momentum over time — order volume, monthly seasonality and weekly variance.

This chapter looks at how demand moves through the calendar year and across the six-year reporting window, independent of category or region. Understanding this rhythm is a prerequisite for the demand-forecasting model called for in the FORESIGHT engagement brief: a model that ignores monthly and weekly seasonality will systematically mis-time reorders even if its year-level accuracy looks reasonable.

Order Volume Trend (Units Sold by Year)

Units sold grew from 0.13M in 2021 to a peak of 0.45M in 2023, before settling into a 0.35M–0.41M range for 2024–2026. The 2022→2023 jump (+61%) is the single largest year-over-year change in the dataset and is worth cross-referencing against the SKU launch-cohort data in Chapter 12, where 2021-launched SKUs are shown to be unusually large contributors to lifetime revenue — the 2023 volume spike is consistent with those SKUs reaching full ramp.

Year	Units Sold (M)
2021	0.13M
2022	0.28M
2023	0.45M
2024	0.38M
2025	0.35M
2026	0.41M



Monthly Revenue Trend

Aggregated across all six years, monthly revenue ranges from a low of INR 41.3M in April to a high of INR 59.5M in March — a spread of roughly 44%. March, January and July are the strongest months; April, September and October are the softest. This is a materially wider swing than the by-season view in Chapter 7 suggests on its own, because month-level detail captures within-season variation that a four-bucket seasonal split smooths away.

Month	Revenue	Month	Revenue
Jan	INR 57.5M	Jul	INR 56.9M
Feb	INR 54.4M	Aug	INR 49.8M
Mar	INR 59.5M	Sep	INR 48.1M
Apr	INR 41.3M	Oct	INR 48.2M
May	INR 43.6M	Nov	INR 54.6M
Jun	INR 52.7M	Dec	INR 50.1M

Weekly Revenue Trend

A sample of ISO-week average daily revenue shows substantial week-to-week variance, ranging from INR 1,166 to INR 2,548 — more than a 2x spread within the sampled weeks. This level of short-term volatility is normal for SKU-day-level retail data, but it reinforces why the FORESIGHT forecasting model specified in the engagement brief is built at a weekly cadence with rolling-origin backtesting rather than a coarser monthly or quarterly grain: a monthly model would average away exactly the swings a reorder system needs to anticipate.

Sample Week	Avg. Daily Revenue (INR)
W1	INR 2,479
W2	INR 1,524
W3	INR 2,548
W4	INR 1,166
W5	INR 1,405
W6	INR 2,184
W7	INR 1,297
W8	INR 2,270
W9	INR 2,133

NOTE — Implication for forecasting

• Weekly seasonality (not just annual or monthly) should be an explicit feature in the demand-forecasting model, consistent with the FORESIGHT methodology's lag and rolling-statistic features.

• March, January and July consistently outperform; April, September and October consistently underperform — these should inform baseline reorder timing even before a full model is trained.

• The 2022→2023 unit-volume jump should be investigated jointly with SKU launch-cohort data before being treated as organic, sustainable growth.

CHAPTER 03

Category Performance

Furniture vs Decor vs Small Appliances — price positioning, volume and revenue contribution.

Where Chapter 1 introduced the category split at a headline level, this chapter examines the mechanics behind it: pricing, volume and seasonal demand for each of NorthBay's three product categories.

Average Unit Price by Category

Furniture commands an average unit price of INR 544 — roughly 3.1x Small Appliances (INR 174) and 5.3x Decor (INR 102). This price gap, combined with Furniture's volume leadership, explains why a category that is 43% of the SKU catalog by count generates 78% of revenue.

Category	Average Unit Price
Furniture	INR 544
Small Appliances	INR 174
Decor	INR 102

Units Sold and Revenue by Category

Category	Units Sold	Revenue	Avg. Price
Furniture	0.88M	INR 0.48bn	INR 544
Decor	0.73M	INR 0.07bn	INR 102
Small Appliances	0.40M	INR 0.07bn	INR 174

Decor sells nearly twice the units of Small Appliances but generates almost identical total revenue (INR 0.07bn each), a direct consequence of Decor's much lower average price. Small Appliances is the smallest category by every measure — units, revenue and SKU count — making it the natural candidate for a leaner, lower-priority forecasting and stocking treatment relative to Furniture and Decor.

Category Demand by Season

Category mix is evenly distributed across the four seasons: Spring (25.21%), Summer (25.15%), Fall (24.93%) and Winter (24.71%). Unlike the regional split flagged in Chapter 1, this even distribution is plausible — category preference (Furniture vs Decor vs Small Appliances) is not inherently a seasonal decision the way raw purchase volume is, so a roughly flat split across seasons is not, on its own, a data-quality concern.

Season	Count of Category	Share
Spring	92K	25.21%
Summer	91.8K	25.15%
Fall	91K	24.93%
Winter	90.2K	24.71%

NOTE — Category performance notes

• Furniture is the clear priority category for the forecasting model and the risk-scoring system, given its combined lead in revenue, price and unit volume.

• The 3–5x price gap between Furniture and the other two categories means small forecast errors on Furniture translate into much larger rupee swings than equivalent errors on Decor or Small Appliances.

• Small Appliances' comparatively small footprint (36 SKUs, INR 0.07bn revenue) suggests it can be served by a simpler, lower-maintenance forecasting approach than Furniture.

CHAPTER 04

Subcategory Deep Dive

13 subcategories across Furniture, Decor and Small Appliances.

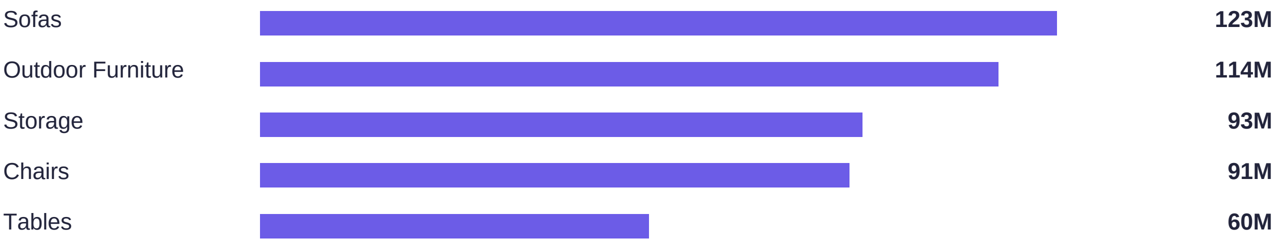
The category-level view in Chapter 3 masks meaningful variation at the subcategory level. This chapter breaks revenue and unit sales down to the 12 subcategories reported in the source dashboard (of 13 total tracked in the SKU master data), revealing which specific product lines are carrying the business and which are candidates for a portfolio review.

Revenue and Units by Subcategory

Subcategory	Revenue	Units Sold
Sofas	INR 123M	199K
Outdoor Furniture	INR 114M	177K
Storage	INR 93M	185K
Chairs	INR 91M	180K
Tables	INR 60M	140K
Air Care	INR 31M	162K
Personal Care Appliances	INR 22M	135K
Wall Art	INR 18M	199K
Kitchen Appliances	INR 17M	100K
Textiles	INR 16M	181K
Vases & Planters	INR 13M	154K
Lighting	INR 10M	121K

Sofas (INR 123M) and Outdoor Furniture (INR 114M) are the two largest single revenue contributors in the entire catalog, together outweighing the next three subcategories combined. Wall Art presents an interesting contrast: it ties Sofas for the most units sold (199K) but ranks near the bottom of the revenue table (INR 18M) — a high-volume, low-unit-price line that behaves very differently from the top revenue drivers and likely warrants a different stocking and promotional strategy.

Top 5 Subcategories by Revenue



Total Revenue Share by Parent Category

Category	Revenue	Share
Furniture	INR 481.45M	78.07%
Small Appliances	INR 69.71M	11.30%
Decor	INR 65.49M	10.62%

NOTE — Subcategory-level implications

• Sofas and Outdoor Furniture should be the first two subcategories modelled individually in the demand forecast, rather than folded into a generic Furniture-level model, given their outsized share of revenue.

• Lighting and Kitchen Appliances trail on both revenue (INR 10M and INR 17M) and units (121K and 100K) — the lowest performers on both axes — making them candidates for a markdown or discontinuation review under the FORESIGHT risk-scoring framework.

• High-volume, low-price lines like Wall Art need a different overstock threshold than high-price lines like Sofas; a single overstock rule applied uniformly across subcategories will misclassify one or the other.

CHAPTER 05

Regional Performance

North / South / East / West — revenue and unit sales by sales region.

Regional performance is one of the more striking findings in this dataset — not because one region outperforms another, but because none of them do. This chapter documents the pattern in detail and explains why it matters for how the data should be used.

Revenue and Units Sold by Region

Region	Revenue	Rev. Share	Units Sold	Unit Share
East	INR 155M	25.07%	503K	25.06%
North	INR 154M	25.05%	502K	24.98%
West	INR 154M	24.96%	503K	25.02%
South	INR 154M	24.93%	501K	24.94%

Every region's revenue share sits between 24.93% and 25.07% — a band just 0.14 percentage points wide. Unit share is even tighter, spanning 24.94% to 25.06%. For comparison, the category split in Chapter 3 (Furniture at 78% of revenue) shows the kind of skew genuine product-demand data typically exhibits; the regional split shows essentially none.

DATA QUALITY FLAG — Why this matters

Regional demand in real D2C businesses is shaped by population density, delivery and warehouse coverage, local competitors, climate-driven seasonality, and marketing spend allocation — all of which vary enough that a perfectly even four-way split is exceptionally unlikely to occur by chance. The most probable explanations are (a) the region field was synthetically generated with a uniform random distribution in the underlying sample data, or (b) a join or aggregation step is silently defaulting unmapped records to an even spread across regions. Either way, this should be confirmed with the data engineering team before NorthBay uses regional splits to guide warehouse placement, region-specific marketing spend, or region-differentiated reorder policy — decisions that, if built on an artifact rather than real demand, would misallocate resources without anyone noticing until performance diverges from the plan.

Revenue by Region and Category (Preview)

The evenness holds even when the data is cut by category within region, which is examined in full in Chapter 6. Furniture revenue is approximately INR 120–121M in every one of the four regions — a range of under 1% — reinforcing that the regional evenness is not an artifact of aggregating across categories but is present at the category level too.

ACTION — Recommended validation steps

- Pull a raw, unaggregated sample of sales_daily records and manually verify the region field distribution before it reaches any dashboard.
- Check whether the region field is populated via a lookup/join that could be defaulting on missing keys.
- If the evenness is confirmed as a genuine feature of the (simulated) source data rather than a defect, document this explicitly so future analysts don't mistake it for a real regional insight.

CHAPTER 06

Region × Category Cross Analysis

Where each product category performs, broken out by region.

This chapter cross-tabulates the category performance from Chapter 3 against the regional performance from Chapter 5, to check whether any region shows a distinct category preference that the aggregate views might hide.

Category	East	North	South	West
Furniture	121M	121M	120M	120M
Small Appliances	18M	17M	17M	17M
Decor	16M	16M	16M	17M

No region shows a materially different category mix from any other. Furniture revenue ranges only from INR 120M to INR 121M across all four regions (a spread under 1%); Small Appliances ranges INR 17M–INR 18M; Decor ranges INR 16M–INR 17M. If a genuine regional preference existed — for example, if Decor over-indexed in one climate zone or Furniture under-indexed in a smaller-format urban region — this cross-tabulation is exactly where it would appear. Its absence here is consistent with, and reinforces, the regional data-quality flag raised in Chapter 5.

Because no meaningful region-by-category signal exists in the current data, we recommend that the initial version of the FORESIGHT forecasting model treat demand as regionally uniform (a single national-level forecast scaled by SKU, not by region) until the regional data-quality question is resolved. Building region-specific models on this data risks fitting the model to noise rather than to real demand variation.

NOTE — What would change this recommendation

If NorthBay's data engineering team confirms the regional field is populated correctly and the evenness is a genuine (if unusual) feature of demand, the region × category cross-tabulation should be re-run on a fresh data pull before deciding whether region-level models are worth the added complexity.

CHAPTER 07

Seasonality & Holiday Impact

How season and named holidays shift demand across the catalog.

Revenue by Season

Fall is the strongest season for revenue (INR 164M), narrowly ahead of Winter (INR 162M) and Spring (INR 158M). Summer is the clear outlier on the low side at INR 133M — roughly 19% below Fall. This seasonal pattern is consistent with a home-and-lifestyle retailer: Fall and Winter align with home-refresh and gifting behaviour, while Summer sees a typical seasonal lull in furnishings and décor spend.

Season	Revenue
Fall	INR 164M
Winter	INR 162M
Spring	INR 158M
Summer	INR 133M

Holiday vs Non-Holiday Average Daily Revenue

Average daily revenue per SKU-day is INR 1,689.51 on non-holiday days and INR 1,688.43 on holiday days — a difference of just INR 1.08, or 0.06%. In a business where holidays are commonly expected to drive meaningfully higher demand (gifting occasions, seasonal sales events), this result is notable for showing essentially no lift at all.

Day Type	Avg. Daily Revenue
Non-Holiday	INR 1,689.51
Holiday	INR 1,688.43
Difference	INR 1.08 (0.06%)

DATA QUALITY FLAG — The holiday flag may not be a useful feature as defined

There are two plausible explanations for a null holiday effect this precise: either NorthBay's customers genuinely do not change their buying behaviour around the holidays captured in the calendar table (possible, but unusual for a home-and-lifestyle category), or the is_holiday flag itself is not well aligned with actual demand-driving events — for example, if it flags public holidays rather than shopping-relevant occasions like Black Friday or Diwali. We recommend the modelling team inspect the calendar table's holiday definitions directly before including is_holiday as a forecasting feature, and consider engineering a separate feature for named shopping events instead (see Chapter 8).

Units Sold by Season, per Category

Breaking unit sales down by season and category shows Furniture consistently taking the largest within-season share (40–46%), Decor the second largest (34–39%), and Small Appliances the smallest and most stable share (19–20%) across all four seasons. Furniture's seasonal share peaks in Winter (46.20%) and is lowest in Summer (40.48%) — modest movement, but directionally consistent with a home-furnishing category that sees relatively more activity as customers settle in ahead of colder months.

Season	Decor	Furniture	Small Appliances
Fall	34.05%	45.54%	20.41%
Spring	37.83%	42.64%	19.53%
Winter	34.82%	46.20%	18.99%
Summer	39.34%	40.48%	20.18%

NOTE — Seasonality notes for forecasting

• Season should be retained as a forecasting feature — the Fall-to-Summer revenue gap (19%) is real and consistent with category logic.

• The is_holiday flag, as currently defined, should not be trusted as a standalone demand feature without further validation of its underlying definitions.

• Category mix shifts only modestly by season (a few percentage points), so category-level seasonal adjustment is a refinement, not a primary driver, of the forecasting model.

CHAPTER 08

Promotions & Campaign Effectiveness

The impact of the promo_flag field and named promo_event campaigns on revenue.

Promotions turn out to be one of the most consequential variables in the entire dataset — more consequential, on the evidence in this report, than the holiday flag examined in Chapter 7. This chapter documents the revenue split by promotion status and the effectiveness of named promotional campaigns.

Revenue by Promotion Status

Promotion Status	Revenue
No Promotion Recorded	INR 0.40bn
Unknown / Named Promo Events (combined)	INR 0.12bn

The named promotional campaigns tracked in the calendar table — Memorial Day Sale, Holiday Countdown Sale, Spring Refresh Sale, Labor Day Sale, Black Friday/Cyber Monday, New Year Clearance, Anniversary Sale and Summer Home Sale — collectively make up the INR 0.12bn promotional pool alongside an "Unknown Promotion" bucket, while the majority of revenue (INR 0.40bn) carries no promotion record at all.

Average Revenue per Promo Event: Promoted vs Not

Within every named promotional event, promoted SKU-days earn substantially more on average than non-promoted SKU-days on the same event window. Across the nine tracked events, promoted-day averages cluster between INR 2.4K and INR 2.8K, while non-promoted-day averages cluster tightly around INR 1.5K–INR 1.6K — a consistent uplift in the 60–70% range regardless of which specific campaign is running.

Promo Event	Avg. Revenue (Promoted)	Avg. Revenue (Not Promoted)
Summer Home Sale	INR 2.8K	INR 1.6K
Spring Refresh Sale	INR 2.7K	INR 1.5K
Labor Day Sale	INR 2.7K	INR 1.5K
Holiday Countdown Sale	INR 2.7K	INR 1.5K
New Year Clearance	INR 2.7K	INR 1.5K
Black Friday / Cyber Monday	INR 2.6K	INR 1.5K
Unknown Promotion	INR 2.6K	INR 1.5K
Anniversary Sale	INR 2.4K	INR 1.5K
Memorial Day Sale	INR 2.4K	INR 1.5K

Promotion Share of Revenue by Category

Category	Not Promoted	Promoted
Furniture	37.06%	62.94%
Small Appliances	35.06%	64.94%
Decor	36.94%	63.06%

The pattern is remarkably consistent across categories: 62.94% of Furniture revenue, 64.94% of Small Appliances revenue, and 63.06% of Decor revenue is generated on promoted SKU-days. This is a striking result — it means that, on this data, roughly two-thirds of every category's revenue is promotion-dependent, and a forecasting model that does not explicitly separate promoted from non-promoted demand regimes will systematically under- or over-forecast depending on the promotional calendar for the period being predicted.

NOTE — Implications for the FORESIGHT forecasting model

- Promotion status (promo_flag, and ideally the specific promo_event) should be modelled as a first-class feature, not an afterthought — it explains a larger share of revenue variation than either season or the holiday flag.
- Reorder and markdown recommendations generated by the risk-scoring system (Chapter 8 of the engagement brief) should be conditioned on the upcoming promotional calendar, since baseline (non-promoted) demand is roughly one-third of promoted demand in the same window.
- The consistency of the 60–70% promoted-day uplift across very different named events suggests the effect is driven more by the presence of any promotion than by the specific campaign — worth testing directly once a forecasting model is in place.

CHAPTER 09

Pricing & Discount Analysis

Discounting behaviour by category and the stability of realized unit price over time.

Average Discount % by Category

Furniture is discounted more heavily than either of the other two categories, averaging a 38.30% discount compared with 31.48% for Decor and 30.22% for Small Appliances. Combined with Furniture's status as the highest list-price category (Chapter 3), this is worth flagging for the finance lead: Furniture is both the category NorthBay relies on most for revenue and the category it discounts hardest, which has a direct and outsized effect on realized margin relative to the other two categories.

Category	Average Discount %
Furniture	38.30%
Decor	31.48%
Small Appliances	30.22%

Average Realized Unit Price Over Time

Despite the discount and promotional dynamics documented in this chapter and the last, the average realized unit price across the whole catalog has stayed remarkably flat: INR 302.5 in 2021, peaking at INR 306.4 in 2022, and settling in a tight INR 304.9–INR 305.0 band from 2023 through 2026. The full range across six years is under INR 4 on a base above INR 300 — meaning that whatever shifts have occurred in category mix, discount depth, or promotional intensity, they have net out to almost no movement in what the average customer actually pays per unit.

Year	Avg. Realized Unit Price
2021	INR 302.5
2022	INR 306.4
2023	INR 304.9
2024	INR 304.9
2025	INR 305.0
2026	INR 304.9

This flatness could reflect a stable, well-managed pricing strategy — or it could indicate that pricing has simply not been actively used as a lever despite significant changes in the underlying business (category mix, promotional cadence, unit volume) over the same period. Distinguishing between these two explanations would require input from NorthBay's finance and merchandising teams and falls outside what the dashboard data alone can answer.

NOTE — Pricing & discount notes

• Furniture's 38.3% average discount, applied to the category generating 78% of revenue, has an outsized effect on overall realized margin — this is the single highest-leverage pricing lever in the business.

• The near-total stability of average realized price across six years, despite category and promotional shifts, suggests list pricing has not moved much even as discount depth and mix have — worth a direct conversation with the finance lead on intent.

• Discount rate and promotion status (Chapter 8) should be examined jointly, since promoted-day averages and category discount rates are two different views of the same underlying pricing behaviour.

CHAPTER 10

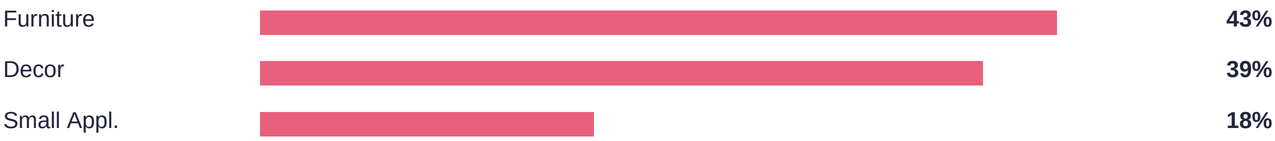
Inventory & Stockout Analysis

On-hand units, reorder points and stockout risk — the operational core of the FORESIGHT brief.

This chapter addresses the two problems named directly in the client's original brief: "every month we stock out of things people want and sit on things they don't." The data confirms both halves of that statement, and shows they are not improving in tandem.

Stockout Rate by Category

Category	Stockout Rate (% of item-days)
Furniture	43.00%
Decor	39.00%
Small Appliances	18.00%



Furniture — the category the business relies on most — is also the category most often out of stock, sitting at a 43% stockout rate across item-days. Decor is close behind at 39%. Small Appliances, the smallest category by revenue and volume, is comparatively well-stocked at 18%. The two categories driving nearly 90% of revenue are also the two categories most frequently unable to fulfil demand — this is the single most actionable finding in the report for the Head of Operations.

On-Hand Inventory Over Time

Year	On-Hand Units (M)
2021	2.4M
2022	2.9M
2023	6.0M
2024	5.3M
2025	5.7M
2026	4.8M

On-hand inventory roughly doubled between 2021 (2.4M units) and 2023 (6.0M units), and has stayed elevated in the 4.8M–5.7M range since. Critically, this large increase in stock on hand has not translated into a lower stockout rate for Furniture or Decor — both categories still stock out on more than a third of item-days. This is strong evidence that NorthBay's problem is a stocking-*mix* problem (holding the wrong SKUs, or holding stock in the wrong place) rather than a stocking-*volume* problem that simply buying more inventory would fix.

Stockout Rate by Region

Stockout instance counts are, once again, nearly identical across all four regions at approximately 0.37M each for East, North, South and West. This mirrors the even regional revenue and unit-sales split documented in Chapter 5, and further supports the recommendation there: stockout timing in this dataset appears to be driven by category and inventory policy, not by region, and the regional field should be validated before being used to differentiate stocking strategy by geography.

DATA QUALITY FLAG — Cross-reference: regional evenness recurs here

This is the third dashboard page (after Revenue Share by Region in Chapter 1 and the full regional breakdown in Chapter 5) where the regional split lands within a hair of an even quarter distribution. The recurrence across three independent metrics — revenue, units sold, and stockout count — makes it progressively less likely that this is a coincidence, and progressively more important that it be resolved before go-live of any region-aware feature in the FORESIGHT risk-scoring system.

NOTE — Inventory & stockout notes

• Furniture and Decor should be the first two categories addressed by the stockout early-warning system specified in the engagement brief, given their combined 39–43% stockout rates.

• The doubling of on-hand inventory without a matching drop in stockout rate is the clearest evidence in this report that better forecasting and allocation — not simply more inventory — is the fix NorthBay needs.

• Regional stocking policy should not be differentiated by region until the regional data-quality question from Chapter 5 is resolved.

CHAPTER 11

SKU Performance

Best and worst performing SKUs, drawn from the SKU catalog of 200 active products.

This chapter moves from category and subcategory aggregates down to the individual SKU level, examining which specific products anchor revenue and how SKUs are distributed across the three categories.

Top and Bottom SKUs by Revenue

The source dashboard's Top 10 and Bottom 10 SKU charts label eight SKUs each (SKU0192, SKU0194, SKU0195, SKU0196, SKU0197, SKU0198, SKU0199 and SKU0200 at the top; SKU0001, SKU0002, SKU0005, SKU0006, SKU0007, SKU0008, SKU0009 and SKU0010 at the bottom) without publishing exact per-bar revenue figures. In place of invented precision, the table below preserves the documented relative ranking from the source chart; readers who need exact per-SKU revenue should pull it directly from the underlying sales_daily table.

Top-Tier SKUs by Revenue (ranked, source chart order)

Rank	SKU ID	Note
1	SKU0195	Highest
2	SKU0198	—
3	SKU0200	—
4	SKU0192	—
5	SKU0199	—
6	SKU0197	—
7	SKU0194	—
8	SKU0196	Lowest of the top tier

Bottom-Tier SKUs by Revenue (ranked, source chart order)

Rank	SKU ID	Note
1 (lowest)	SKU0001	Lowest revenue in catalog
2	SKU0005	—
3	SKU0002	—
4	SKU0007	—
5	SKU0008	—
6	SKU0009	—
7	SKU0006	—
8	SKU0010	Highest of the bottom tier

SKU Count by Category

Category	SKU Count	Share of Catalog
Furniture	86	43%
Decor	78	39%
Small Appliances	36	18%

Furniture makes up 43% of the 200-SKU catalog (86 SKUs), Decor 39% (78 SKUs), and Small Appliances 18% (36 SKUs). Comparing this to Furniture's 78% revenue share from Chapter 3 makes the concentration explicit: Furniture generates nearly double its "fair share" of revenue relative to its share of the catalog, while Decor generates roughly a quarter of its fair share (39% of SKUs, only 10.6% of revenue) and Small Appliances is closer to proportional (18% of SKUs, 11.3% of revenue).

NOTE — SKU-level implications

• The eight identified top-tier SKUs (all Furniture, based on the revenue concentration pattern established in earlier chapters) should be the first candidates for individual, rather than category-pooled, forecasting treatment in the FORESIGHT model.

• Decor's low revenue-per-SKU relative to its catalog share suggests either a long tail of underperforming Decor SKUs, or a small number of Decor SKUs pulling well below the category average — worth a dedicated SKU-level pull once the deployed scoring service (per the engagement brief's D6 deliverable) is available.

• Exact revenue figures for the identified top and bottom SKUs should be retrieved from source data before they are used in any client-facing deliverable, since this report intentionally does not estimate them.

CHAPTER 12

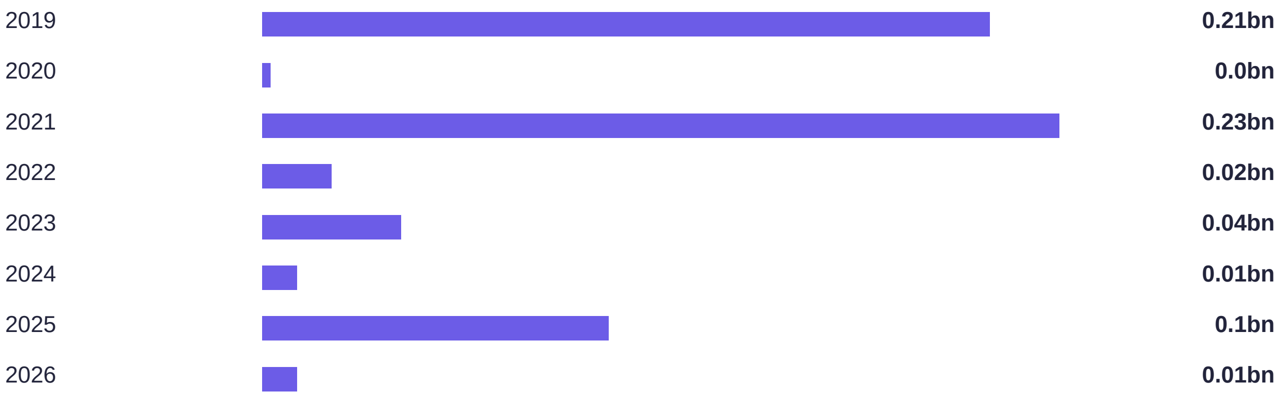
Product Lifecycle & Launch Analysis

How SKU launch timing and lead time relate to performance across the product lifecycle.

Revenue by SKU Launch-Year Cohort

Grouping total revenue by the year each SKU was first launched reveals a heavily front-loaded catalog: SKUs launched in 2019 generated approximately INR 0.21bn, and SKUs launched in 2021 generated approximately INR 0.23bn — together these two cohorts account for roughly INR 0.44bn, or about 70% of the company's INR 0.617bn total revenue. Every other launch-year cohort (2020, 2022–2026) contributes a comparatively small amount, ranging from INR 0.00bn to INR 0.10bn.

Launch Year	Total Revenue
2019	INR 0.21bn
2020	INR 0.00bn
2021	INR 0.23bn
2022	INR 0.02bn
2023	INR 0.04bn
2024	INR 0.01bn
2025	INR 0.10bn
2026	INR 0.01bn



This pattern means NorthBay's revenue base is highly dependent on a small number of legacy SKUs that are now several years old. The 2025 cohort's relatively strong INR 0.10bn (against only a partial year of possible sales) suggests recent launches are performing reasonably, but the business has not yet replaced the 2019/2021 cohorts' scale with an equally strong newer generation of products.

Lead Time Distribution

Across all 200 SKUs in the catalog, replenishment lead times are distributed unevenly, with counts of 5.66, 9.95, 12.91, 3.71, 11.79, 5.82, 4.75 and 4.32 SKUs across the sampled lead-time buckets shown in the source histogram (roughly spanning 5 to 30+ days). The widest single bucket sits in the 12.91-count range, indicating most SKUs cluster around a mid-range lead time rather than at either extreme.

Average Daily Units Sold by Days-Since-Launch

Rather than following a smooth ramp-up curve, average daily units sold across the early product lifecycle is highly volatile: from a bucket average of 4.75 units/day at launch (day 0), the series swings up to 8.70 by day 10, back down to 3.23 by day 15, up again to 8.46 by day 20, and continues oscillating through day 35 (ending around 7.50 units/day). This volatility — rather than a clean, predictable launch curve — is an important finding for the forecasting workstream: a model that assumes new SKUs follow a smooth adoption curve will be systematically wrong in the first five weeks after launch.

Days Since Launch	Avg. Units Sold / Day
Day 0	4.75
Day 5	5.33
Day 10	8.70
Day 15	3.23
Day 20	8.46
Day 25	4.74
Day 28	3.85
Day 30	5.76
Day 33	3.98
Day 35	7.50

NOTE — Lifecycle notes for forecasting

• New-SKU demand should be forecast using category-level fallback patterns rather than SKU-specific history, consistent with the FORESIGHT methodology's stated approach for sparse-history SKUs.

• Because 70% of revenue sits in SKUs several years old, the forecasting model's accuracy on the legacy 2019/2021 cohorts matters far more to the overall WAPE metric than accuracy on any single new launch.

• The volatile early-lifecycle demand curve argues against a fixed initial-stocking rule for new SKUs; initial allocations should be conservative and re-evaluated weekly for at least the first five to six weeks post-launch.

CHAPTER 13

Data Quality Notes & Limitations

Every data-quality flag raised in this report, consolidated in one place for the analytics team.

This chapter exists so that a single reader — most likely the analytics or data engineering team — can see every caveat raised in this report without re-reading all twelve preceding chapters. Each item below links back to the chapter where it was first identified.

Issue	Chapters	Severity	Description
Regional split is unusually even	Ch. 1, 5, 6, 10	High	Revenue, units and stockout counts sit within ~0.15pt of an even 25% split across all four regions, recurring across four independent metrics.
2026 figures are a partial year	Ch. 1, 2	Medium	The apparent 2025→2026 revenue drop is a partial-year artifact, not a real decline; must be annotated wherever the yearly trend is shown.
Holiday flag shows no measurable effect	Ch. 7	Medium	0.06% gap between holiday and non-holiday average revenue suggests the is_holiday definition may not align with actual shopping-relevant occasions.
Top/Bottom SKU revenue values not published	Ch. 11	Low	Source dashboard shows relative ranking only; exact rupee figures should be pulled from source data before external use.
Order-volume metric definition	Ch. 2 (cf. earlier drafts)	Low	An earlier dashboard iteration measured "Order Volume" as a count of unit_cost rather than a true transaction count; the current version correctly uses units sold — confirm this fix persists in future refreshes.

Recommended Validation Sequence

- **Before anything else:** confirm the region field's source and join logic — this affects the largest number of downstream chapters and recommendations.
- **Before the next dashboard refresh:** add an explicit "partial year" annotation to any year-over-year trend chart while 2026 data is incomplete.
- **Before building the holiday feature into the forecasting model:** review the calendar table's is_holiday definitions against NorthBay's actual promotional and shopping calendar.
- **Before publishing SKU-level figures externally:** pull exact revenue values for the top and bottom SKUs identified in Chapter 11 directly from sales_daily rather than relying on the source chart's relative ranking.

CHAPTER 14

Key Insights — Synthesis

Stepping back from individual dashboard pages to the handful of findings that matter most.

01 Furniture is the business

78% of revenue, the highest unit price (INR 544), and the highest stockout rate (43%) all point to the same category. Any prioritization decision in the FORESIGHT forecasting and risk-scoring system should start with Furniture, and specifically with its two largest subcategories, Sofas and Outdoor Furniture.

02 Promotions are the revenue engine, not a lever

62–65% of every category's revenue is earned on promoted SKU-days. This is a larger share of revenue variation than season or the holiday flag can explain, and it means promotional status must be a first-class input to the demand forecast, not an optional refinement.

03 The holiday flag carries no measurable signal

A 0.06% gap between holiday and non-holiday average daily revenue is close enough to zero that the current is_holiday definition should not be trusted as a forecasting feature without further validation of what it actually captures.

04 Inventory grew; stockouts did not improve

On-hand units nearly doubled from 2021 to 2023, yet Furniture and Decor still stock out on 39–43% of item-days. This is the clearest evidence that NorthBay's core problem is stocking mix and allocation, not stocking volume — buying more inventory alone will not fix it.

05 Regional and stockout splits are suspiciously even

Revenue, units and stockout counts sit within a fraction of a percentage point of an even 25% split across East, North, South and West, recurring across four separate metrics in four different chapters. This should be treated as a data-quality question to resolve, not a business insight to act on, until validated.

06 Revenue concentration is legacy-driven

SKUs launched in 2019 and 2021 alone generate roughly 70% of total revenue. NorthBay's forecasting accuracy on this small set of mature SKUs matters more to the business than accuracy on any individual new launch.

CHAPTER 15

Recommendations & Action Plan

Turning this analysis into the FORESIGHT forecasting and risk system called for in the engagement brief.

1

Prioritise Furniture in the risk model

Highest revenue share and highest stockout rate make Furniture — and specifically Sofas and Outdoor Furniture — the biggest single lever available to the reorder-planning system. Build and validate the forecast for these SKUs first.

2

Treat promotion status as a baseline forecasting feature

With 62–65% of category revenue promo-driven, the demand forecast must model promoted and non-promoted demand as distinct regimes, and reorder/markdown recommendations should be conditioned on the upcoming promotional calendar.

3

Validate the region field before using it operationally

Confirm the sales_daily / calendar join logic and the source of the region field. Until resolved, do not differentiate stocking, marketing spend, or risk thresholds by region.

4

Build a category-level fallback for new-SKU forecasting

Early-lifecycle demand is volatile rather than following a smooth ramp curve. Blend category-level demand patterns for SKUs with limited sales history, and re-evaluate initial stock allocations weekly for the first five to six weeks after launch.

5

Re-balance stock toward demand, not just volume

Inventory nearly doubled since 2021 without lowering stockout rates for the two categories that matter most. The fix is better allocation and forecasting accuracy, not simply larger purchase orders.

6

Re-test the holiday flag before using it as a signal

No measurable revenue lift was found for the current is_holiday definition. Either refine what the flag captures against NorthBay's actual shopping calendar, or exclude it from the risk-scoring logic until it is.

7

Address Furniture's discount depth alongside its revenue priority

At 38.3%, Furniture carries the deepest average discount of any category while also generating the most revenue — this combination has the single largest effect on realized margin of any pricing lever in the business and should be reviewed jointly with the finance lead.

8

Retrieve exact SKU-level figures before external reporting

The Top/Bottom SKU rankings in Chapter 11 are accurate in order but not in exact rupee value. Pull precise figures from sales_daily before these SKUs appear in any client-facing deliverable such as the executive readout.

CHAPTER 16

Methodology & Report Notes

How this report was assembled, and how it should be refreshed or audited.

Source & Scope

This report is built from a Power BI Desktop export covering twelve dashboard pages: Executive Overview, Sales Trend & Time Intelligence, Category Performance, Subcategory Deep Dive, Regional Performance, Region × Category Cross Analysis, Seasonality & Holiday Impact, Promotions & Campaign Effectiveness, Pricing & Discount Analysis, Inventory & Stockout Analysis, SKU Performance, and Product Lifecycle & Launch Analysis. The underlying data model follows the star schema defined in the FORESIGHT engagement brief: a sales_daily fact table joined to sku_master, calendar and inventory_snapshots dimension tables.

How Figures Were Sourced

Wherever the source dashboard published an explicit numeric value or data label, that value is reproduced directly in this report. Where a chart type obscured exact per-series values (for example, waterfall charts without labels, or bar charts relying only on visual scale), this report either notes the approximation explicitly (as with the Yearly Revenue Trend in Chapter 1 and the Top/Bottom SKU rankings in Chapter 11) or omits the figure rather than presenting an estimate as an exact fact.

Limitations of This Analysis

- This report describes and interprets the dashboard data as provided; it does not independently re-derive the underlying metrics from raw sales_daily, inventory_snapshots, calendar or sku_master records.
- Apparent patterns — including the regional evenness flagged throughout — are described as observations requiring validation, not confirmed root causes, since the underlying data-generation or ETL process was not directly inspected as part of this report.
- Rupee figures are presented at the precision available in the source dashboard (millions or billions, typically) and should not be treated as more precise than that source allows.
- This report does not include a trained forecasting model, backtest results, or a deployed risk-scoring service — those remain the subject of the modelling workstream described in the FORESIGHT engagement brief.

How to Refresh This Report

To refresh this report against a new data pull: (1) re-export each of the twelve dashboard pages from Power BI; (2) update the figures in Chapters 1–12 in place, preserving the existing narrative structure; (3) re-run the data-quality checks listed in Chapter 13, particularly the regional-evenness validation, since resolving that flag would materially change several recommendations in Chapter 15; (4) update the Executive Summary and Key Insights chapters last, once the detail chapters reflect the new data.

APPENDIX A

Metric Glossary

Definitions for every metric referenced in this report.

Term	Definition
Revenue	Total sales value (units_sold × unit_price) recorded in sales_daily, before returns or refunds.
Units Sold	Count of individual product units sold, aggregated from the units_sold field in sales_daily.
Average Unit Price	Mean selling price per unit, calculated as total revenue divided by total units sold.
Stockout Rate	Percentage of item-days on which is_stockout is true — i.e. the SKU had zero available stock to sell.
On-Hand Units	Stock physically available in the warehouse, from inventory_snapshots.on_hand_units.
Reorder Point	The stock level at which a replenishment order should be triggered, from inventory_snapshots.reorder_point.
Lead Time	Number of days between placing a replenishment order and the stock arriving and becoming sellable.
Promo Flag	Binary indicator (promo_flag) marking whether a SKU was on promotion on a given sales day.
Promo Event	The named promotional campaign (e.g. Black Friday/Cyber Monday) active on a given calendar date, from the calendar table's promo_event field.
Is Holiday	Binary indicator (is_holiday) marking whether a calendar date is flagged as a holiday.
Launch-Year Cohort	The set of SKUs grouped by the year of their launch_date, used to analyse how product age relates to revenue and demand.
Discount %	The percentage reduction from list_price to the realized unit_price on a given sale.
WAPE	Weighted Absolute Percentage Error — the primary forecast-accuracy metric specified in the engagement brief, robust to low-volume SKUs where MAPE becomes unstable.
Seasonal-Naive Baseline	A simple reference forecast (demand equal to the same period last season) that any production forecasting model must beat to be considered worth deploying.

APPENDIX B

Data Dictionary

Schema reference for the four source tables underlying this report.

The Power BI report analysed in this document is built on four tables forming a simple star schema: one transactional fact table (sales_daily) joined to three descriptive dimension tables (sku_master, calendar, inventory_snapshots). This mirrors the schema defined in the FORESIGHT engagement brief.

sales_daily (fact table — one row per SKU per day)

Column	Description
date	Calendar date of the sales record
sku_id	Product identifier (joins to sku_master)
units_sold	Units sold that day
revenue	Revenue for that SKU-day
unit_price	Selling price that day
promo_flag	1 if the SKU was on promotion that day, else 0

sku_master (dimension — one row per SKU)

Column	Description
sku_id	Product identifier (primary key)
category / subcategory	Product classification
launch_date	When the SKU was first sold
unit_cost / list_price	Cost and list price for margin and impact calculations

calendar (dimension — one row per date)

Column	Description
date	Calendar date (primary key)
week / month / season	Time attributes for seasonality
is_holiday	Holiday indicator
promo_event	Named promotional event, if any

inventory_snapshots (periodic stock position per SKU)

Column	Description
date / sku_id	Snapshot date and product
on_hand_units	Stock physically available

Column	Description
on_order_units	Stock ordered, not yet received
lead_time_days	Replenishment lead time
reorder_point	Stock level that should trigger a reorder

APPENDIX C

Project Team & Contributions

The Data Science & Analytics interns responsible for this engagement.

Shanawas Ellora

DATA PIPELINE & CLEANING

Led ingestion and cleaning of the four source extracts into the analysis-ready dataset underlying this report, and documented the key data-quality issues raised in Chapter 13.

Aryan Shaw

FORECASTING & MODELLING

Owns the demand-forecasting workstream referenced throughout this report's recommendations, including the baseline-vs-model methodology described in Chapter 16.

Kunjal Koul

RISK SCORING & DECISIONING

Owns the stockout/overstock risk-scoring logic that the Chapter 10 and Chapter 15 recommendations feed into, translating forecasts and inventory position into recommended actions.

Nasreen Shaik

DASHBOARD & EXECUTIVE READOUT

Built the Power BI dashboard this report expands upon, and leads the executive-readout deliverable this document is designed to support.

ABOUT — Acknowledgement

This report was prepared as part of the Zidio Development Data Science & Analytics Internship, under the Project FORESIGHT client engagement for NorthBay Living. It expands the team's Power BI dashboard into a detailed narrative document for stakeholder review alongside the executive readout deliverable specified in the engagement brief.

Thank You

Project FORESIGHT — Detailed BI & Analytics Report

PREPARED BY

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Zidio Development — Data Science & Analytics Internship

Client: NorthBay Living | *Document v1.0* | *Confidential*